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SPONGE

Abstract

A cleaning device comprises an elongated sponge body with a first end, a second end, and an outer surface connecting the ends. A longitudinal axis traverses through center points of each end. The device features a recessed portion of the outer surface extending between the ends, defining at least one recess wall oriented radially away from the longitudinal axis. This design allows for cleaning liquid spills and curved surfaces. The device may have a tapered shape and a central bore. The recessed portion may have two opposing recess walls. The body may be circular in cross-section except for the recessed portion. The outer surface may include curved and flat portions. Caps may seal open ends. Abrading layers may be included on portions of the outer and inner surfaces. The device can be used vertically or horizontally, with protruding edges acting as a squeegee for various surfaces.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application 63/630,725, filed on Feb. 26, 2024, and incorporated herein by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH AND DEVELOPMENT [0002] Not Applicable.

FIELD OF THE INVENTION

[0003] The present invention relates to cleaning devices, particularly to a multifunctional sponge with a tubular, hollow design that provides versatile cleaning capabilities for various surfaces and objects, including hard-to-reach areas and curved surfaces.

BACKGROUND

[0004] Cleaning sponges are widely used for various household and commercial cleaning tasks. Current cleaning sponges work well for surfaces they can reach. However, these sponges often have difficulty accessing many areas and objects that need cleaning, such as doorknobs, door handles, and the entirety of utensils and other elongated items.

[0005] Existing sponges and scrubbing pads are typically rectangular, square, or round with flat surfaces on top and bottom. Some have holes or openings, but these aren't tubular, long, and hollow, and the openings aren't large enough to fit over objects like doorknobs. The typical sponges on the market today range in size from two inches thick and up to four inches in height.

[0006] These conventional sponges pose problems when cleaning narrow and deep dishes, glasses, cups, and vases. They also struggle with curved edges, bullnose counter tops, and round objects. Without having to press one's hand into the object, it can be difficult or impossible to reach the bottom of tall drinking glasses and vases with current sponges.

[0007] The drawbacks of existing sponges include their inability to effectively clean around curved edges, corners on walls, counter top and table top edges, stair railings, moldings, and curved furniture. They also lack a built-in squeegee feature, making them ineffective at cleaning food and liquid spills off countertops and into sinks. Additionally, current sponges cannot be cleaned from the inside out, leading to trapped food particles and unpleasant odors. There is a need for a cleaning device that can reach difficult areas, conform to curved surfaces, and provide multiple cleaning functions in a single tool. Such a device should be easy to use, protect the user's hands from harsh cleaners, and be effective for a wide range of cleaning tasks. The present invention accomplishes these objectives.

SUMMARY OF THE INVENTION

[0008] A cleaning device is disclosed comprising an elongated sponge body with a first end, a second end, and an outer surface connecting the ends. A longitudinal axis traverses through center points of each end. The device features a recessed portion of the outer surface extending between the ends, defining at least one recess wall oriented radially away from the longitudinal axis. This design allows for cleaning liquid spills and curved surfaces.

[0009] The device may have a tapered shape, with the first end larger than the second. A central bore may extend from the first end towards or to the second end. The recessed portion may have two opposing recess walls. The body may be generally circular in cross-section but for the recessed portion.

[0010] The outer surface may include curved and flat portions. Caps may be provided to seal open ends. Abrading layers may be included on portions of the outer and inner surfaces.

[0011] The device can be used vertically or horizontally, with the user's thumb in the central bore

and fingers in the recessed portion. The protruding edges formed by the recessed portion act as a squeegee for various surfaces and can reach narrow spaces.

[0012] The cleaning device combines curved, recessed, and flat surfaces for versatile cleaning applications. It can be made rigid or semi-rigid to suit different cleaning tasks.

[0013] The present invention addresses the drawbacks of the prior art by providing a multifunctional sponge with a tubular, hollow design. This innovative cleaning device effectively cleans around curved edges, corners, and various objects due to its flexible, recessed outer portion. It incorporates a built-in squeegee feature for efficient liquid removal and can be cleaned from both inside and out, preventing odor and mold buildup. The device's unique shape allows it to reach difficult areas while protecting the user's hands from harsh cleaners. By combining multiple cleaning functions in a single tool, this invention offers a versatile and efficient solution for a wide range of cleaning tasks, overcoming the limitations of conventional sponges and cleaning implements. Other features and advantages of the present invention will become apparent from the following more detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the principles of the invention.

Description

DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a front perspective view of a preferred embodiment of the invention;

[0015] FIG. 2 is a front elevational view of the invention;

[0016] FIG. 3 is a top plan view of the invention;

[0017] FIG. 4 is a bottom plan view of the invention;

[0018] FIG. 5 is a cross-sectional view of another embodiment of the invention;

[0019] FIG. 6 is a top plan view of an alternate embodiment of the invention;

[0020] FIG. 7 is a top plan view of yet another alternate embodiment of the invention; and

[0021] FIG. 8 is a front perspective view of an embodiment having an ergonomic hand-fitting outer surface.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0022] Illustrative embodiments of the invention are described below. The following explanation provides specific details for a thorough understanding of and enabling description for these embodiments. One skilled in the art will understand that the invention may be practiced without such details. In other instances, well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments.

[0023] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.”

Words using the singular or plural number also include the plural or singular number respectively. Additionally, the words “herein,” “above,” “below” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. When the claims use the word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list and any combination of the items in the list. When the word “each” is used to refer to an element that was previously introduced as being at least one in number, the word “each” does not necessarily imply a plurality of the elements, but can also mean a singular element.

[0024] FIGS. 1 and 2 illustrate a cleaning device 10 comprising an elongated sponge body 20 having a first end 22, a second end 28, and an outer surface 25 connecting the first end 22 with the second end 28. A longitudinal axis L.sub.1 of the elongated sponge body 20 traverses through center points C.sub.1, C.sub.2 of each of the first and second ends 22, 28. The elongated sponge

body **20** may be made of materials such as polymers, synthetic urethane sponge, closed cell foam, open cell foam, reticulated foam, polyvinyl alcohol (PVA), natural sponge, loofah, melamine foam, cellulose sponge, or the like. While the outer surface **25** is preferably smooth, it can also be made with ripples, indented points, grooves, or interruptions (not shown) for additional scrubbing effectiveness.

[0025] A recessed portion **30** of the outer surface **25** extends between the first end **22** and the second end **28**. Such a recessed portion **30** defines at least one recess wall **40** (FIG. 4) oriented substantially radially away from the longitudinal axis L.sub.1 of the elongated sponge body **20**. The at least one recess wall **40** can be used to clean liquid spills from a surface, for example, and the recessed portion **30** can be used to clean a curved surface by compressing the elongated sponge body **20** around the curved surface.

[0026] A dimension D.sub.1 of the first end **22** may be larger than a corresponding dimension D.sub.2 of the second end **28**, such that the outer surface **25** of the elongated sponge body **20** is tapered. This tapered design allows for easier insertion into narrow spaces and provides varying surface areas for different cleaning tasks.

[0027] The first end **22** may be open to a central bore **50** extending along the longitudinal axis L.sub.1 of the elongated sponge body **20** towards the second end **28**. Such a central bore **50** defines an inner surface **55** of the elongated sponge body **20**. The central bore **50** may extend fully to the second end **28**, with both the first end **22** and the second end **28** being open to the central bore **50**, as shown in FIG. 5. This design allows for insertion of objects to be cleaned, such as utensils, doorknobs, or handles.

[0028] The recessed portion **30** of the outer surface **25** may define two opposing recess walls **40**, as illustrated in FIG. 3. Such opposing recess walls **40** provide additional cleaning surfaces and allow for better conformity to curved surfaces. A user may insert his thumb through the central bore **50** at either the first end **22** or the second end **28** while wrapping his hand around the outer surface **25**. Curling his fingers around one of the recess walls **40** provides a strong grip of the cleaning device **10** during use.

[0029] As seen in FIG. 6, the elongated sponge body **20** may be generally circular in cross-section across the longitudinal axis L.sub.1, except for the recessed portion **30**. The recessed portion **30** of the outer surface **25** may be a first radial distance R.sub.1 from the longitudinal axis L.sub.1, while a remaining portion of the outer surface **25** is a second radial distance R2 from the longitudinal axis L.sub.1, with the second radial distance R.sub.2 being larger than the first radial distance R.sub.1.

[0030] At least a portion P.sub.1 (FIG. 3) of the outer surface **25** of the elongated sponge body **20** may be curved, while at least another portion P.sub.2 of the outer surface **25** may be flat. These varying surface textures provide versatility for different cleaning tasks.

[0031] A cap **60** (FIG. 5) may be selectively engageable with the first open end **22** of the elongated sponge body **20**, allowing the first open end **22** to be selectively sealed. Similarly, another of the caps **60** may be selectively engageable with the second open end **28** if it is also open. With the caps **60** engaged with the elongated sponge body **20**, dry concentrates or fluids, such as cleaning fluids, sanitizers, deodorizers, or the like, can be held within the central bore **50** during cleaning. Such caps **60** may be made of materials like plastic, rubber, silicone, the same sponge material as the elongated sponge body **20**, or the like, and is preferably held within the central bore **50** by friction.

[0032] At least a portion P.sub.3 of the outer surface **25** may include an abrading layer **70** (FIG. 3), such as a sandpaper type material. Similarly, at least a portion P.sub.4 of the inner surface **55** may include an abrading layer **70**. Such abrading layers **70** may be made of materials like nylon fibers, abrasive particles, textured polymers, or the like, and be used for sanding and polishing a workpiece, for example.

[0033] The cleaning device **10** can be used vertically or horizontally, allowing for versatile cleaning applications. When used horizontally, a user can place their thumb into the central bore **50** at either end of the elongated sponge body **20** and place their three middle fingers in the recessed portion **30**

on the outside of the elongated sponge body **20**, protecting the user's fingers while allowing maximum force to be applied to the cleaning task.

[0034] The protruding edges created by the recessed portion **30** form a squeegee that is ideal for removing water from shower doors, tile walls, kitchen countertops, windows, and glass tabletops. These protruding edges are also useful for reaching narrow edges, grooves in cabinet doors, cracks, and tight edges.

[0035] The cleaning device **10** may be designed with a combination of curved, recessed, flat, and ergonomic outer surfaces (**80**), which can be used when cleaning, scrubbing, disinfecting, sanding, polishing, or sanitizing. The elongated sponge body **20** can be made rigid or semi-rigid with various configurations to suit different cleaning tasks.

[0036] While a particular form of the invention has been illustrated and described, it will be apparent that various modifications can be made without departing from the spirit and scope of the invention. For example, the elongated sponge body **20** may be made in various shapes as desired, such as elongated square, triangular, cylindrical, oval, hexagon, rectangular, diamond, star, pentagon, trapezoid, heart, octagonal (FIG. 7), or the like. Accordingly, it is not intended that the invention be limited, except as by the appended claims.

[0037] Particular terminology used when describing certain features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the invention to the specific embodiments disclosed in the specification, unless the above Detailed Description section explicitly defines such terms. Accordingly, the actual scope of the invention encompasses not only the disclosed embodiments, but also all equivalent ways of practicing or implementing the invention.

[0038] The above detailed description of the embodiments of the invention is not intended to be exhaustive or to limit the invention to the precise form disclosed above or to the particular field of usage mentioned in this disclosure. While specific embodiments of, and examples for, the invention are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. Also, the teachings of the invention provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various embodiments described above can be combined to provide further embodiments.

[0039] All of the above patents and applications and other references, including any that may be listed in accompanying filing papers, are incorporated herein by reference. Aspects of the invention can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further embodiments of the invention.

[0040] Changes can be made to the invention in light of the above "Detailed Description." While the above description details certain embodiments of the invention and describes the best mode contemplated, no matter how detailed the above appears in text, the invention can be practiced in many ways. Therefore, implementation details may vary considerably while still being encompassed by the invention disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated.

[0041] While certain aspects of the invention are presented below in certain claim forms, the inventor contemplates the various aspects of the invention in any number of claim forms. Accordingly, the inventor reserves the right to add additional claims after filing the application to pursue such additional claim forms for other aspects of the invention.

Claims

1. A cleaning device (10) comprising: an elongated sponge body (20) having a first end (22), a second end (28), and an outer surface (25) connecting the first end (22) with the second end (28), a longitudinal axis (L1) of the elongated sponge body (20) traversing through center points (C1,C2) of each of the first (22) and second ends (28); a recessed portion (30) of the outer surface (25) extending between the first end (22) and the second end (28), the recessed portion (30) defining at least one recess wall (40) oriented substantially radially away from the longitudinal axis (L1) of the elongated sponge body (20); wherein the at least one recess wall (40) can be used to clean liquid spills from a surface, and wherein the recessed portion (30) can be used to clean a curved surface by compressing the elongated sponge body (20) around the curved surface.
2. The cleaning device (10) of claim 1 wherein a dimension (D1) of the first end (22) is larger than a corresponding dimension (D2) of the second end (28), such that the outer surface (25) of the elongated sponge body (20) is tapered.
3. The cleaning device (10) of claim 1 wherein the first end (22) is open to a central bore (50) extending along the longitudinal axis (L1) of the elongated sponge body (20) towards the second end (28), the central bore (50) defining an inner surface (55) of the elongated sponge body (20).
4. The cleaning device (10) of claim 3 wherein the central bore (50) extends along the longitudinal axis (L1) of the elongated sponge body (20) to the second end (28), the first end (22) and the second end (28) both being open to the central bore (50).
5. The cleaning device (10) of claim 1 wherein the recessed portion (30) of the outer surface (25) defines two opposing recess walls (40).
6. The cleaning device (10) of claim 1 wherein the elongated sponge body (20) is circular in cross-section across the longitudinal axis (L1), but for the recessed portion (30).
7. The cleaning device (10) of claim 6 wherein the recessed portion (30) of the outer surface (25) is a first radial distance (R1) from the longitudinal axis (L1), and wherein a remaining portion of the outer surface (25) is a second radial distance (R2) from the longitudinal axis (L1), the second radial distance (R2) being larger than the first radial distance (R1).
8. The cleaning device (10) of claim 1 wherein at least a portion (P1) of the outer surface (25) of the elongated sponge body (20) is curved.
9. The cleaning device (10) of claim 1 wherein at least a portion (P2) of the outer surface (25) of the elongated sponge body (20) is flat.
10. The cleaning device (10) of claim 3 further including a cap (60) selectively engageable with the first open end (22) of the elongated sponge body (20), the first open end (22) being selectively sealable with the cap (60).
11. The cleaning device (10) of claim 4 further including a cap (60) selectively engageable with the second open end (28) of the elongated sponge body (20), the second open end (28) being selectively sealable with the cap (60).
12. The cleaning device (10) of claim 1 wherein at least a portion (P3) of the outer surface (25) includes an abrading layer (70).
13. The cleaning device (10) of claim 3 wherein at least a portion (P4) of the inner surface (55) includes an abrading layer (70).
14. A cleaning device (10) comprising: an elongated sponge body (20) having a first end (22), a second end (28), and an outer surface (25) connecting the first end (22) with the second end (28), a longitudinal axis (L1) of the elongated sponge body (20) traversing through center points (C1,C2) of each of the first (22) and second ends (28); wherein a dimension (D1) of the first end (22) is larger than a corresponding dimension (D2) of the second end (28), such that the outer surface (25) of the elongated sponge body (20) is tapered; a recessed portion (30) of the outer surface (25) extending between the first end (22) and the second end (28), the recessed portion (30) defining

two opposing recess walls (40) oriented substantially radially away from the longitudinal axis (L1) of the elongated sponge body (20); wherein the first end (22) is open to a central bore (50) extending along the longitudinal axis (L1) of the elongated sponge body (20) to the second end (28), the central bore (50) defining an inner surface (55) of the elongated sponge body (20), and the first end (22) and the second end (28) both being open to the central bore (50); wherein the elongated sponge body (20) is circular in cross-section across the longitudinal axis (L1), but for the recessed portion (30); wherein the recessed portion (30) of the outer surface (25) is a first radial distance (R1) from the longitudinal axis (L1), and wherein a remaining portion of the outer surface (25) is a second radial distance (R2) from the longitudinal axis (L1), the second radial distance (R2) being larger than the first radial distance (R1); wherein at least a portion (P1) of the outer surface (25) of the elongated sponge body (20) is curved; wherein the opposing recess walls (40) can be used to clean liquid spills from a surface, and wherein the recessed portion (30) can be used to clean a curved surface by compressing the elongated sponge body (20) around the curved surface.
